INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY **9647/03**

Paper 3 Free Response

20 September 2013

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **4 out of 5** questions.
Begin each answer on a fresh sheet of paper.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

This document consists of **13** printed pages and **1** blank page.

Answer **4 out of 5** questions.

1 **(a)** A possible source of energy for the road vehicles of the future is hydrogen. One of the problems still to be solved is the storage of the hydrogen in the vehicle. A conventional tank holding liquid hydrogen would have to be pressurised and refrigerated. In a crash, this type of tank could break resulting in the rapid release of hydrogen and an explosion.

One alternative is to use a fuel tank packed with carbon nanotubes. The hydrogen in the tank would be adsorbed onto the surface of the nanotubes and the pressure in the fuel tank is due to free gaseous hydrogen.

(i) What forces could be responsible for holding the hydrogen on the surface of the nanotubes? Explain your answer.

(ii) An equilibrium exists between the hydrogen gas adsorbed on the surface of the carbon nanotubes and free gaseous hydrogen.

$$H_{2\ adsorbed} \rightleftharpoons H_{2\ gaseous}$$

When a nanotube-packed fuel tank is full of hydrogen there is a steady pressure of hydrogen in the tank.

While hydrogen gas is being removed from the fuel tank to power the car, the pressure in the fuel tank drops very little for some time.

Suggest an explanation for this phenomenon.

[4]

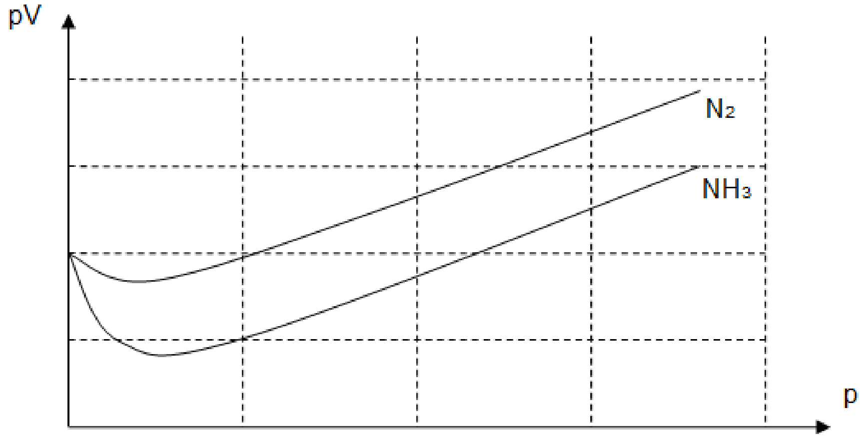
(b) **(i)** In the industry, hydrogen is used in the Haber process to synthesise ammonia as shown below.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

The K_p of the Haber process is 1.34 x 10^{-4} atm^{-2} at 472 °C. A mixture contains 1 mol of nitrogen and 3 mol of hydrogen initially. When it is allowed to reach equilibrium at this temperature, the equilibrium mixture is found to contain 1.5 mol of NH_3. Calculate the total pressure of the system at equilibrium.

(ii) From an experiment, it is found that in organic solvent, ammonia react with an equimolar amount of BF_3 to form a product with a melting point of 260 °C. Explain why these two molecules react in the given manner.

(iii) The following graphs show how the same amount of nitrogen gas and ammonia gas behave over a range of pressure, at the same temperature.



Explain the shape of the graph for ammonia gas.

[5]

(b) Organometallic compounds, usually a metal attached to an R group, can be used to convert carbonyl compounds to alcohols. An example is shown below.

$$R{-}CH{=}O \; + \; R'{-}Li \xrightarrow{H_2O} R{-}CH(OH){-}R' \; + LiOH$$

(i) 2-methyl-butan-2-ol is a specialty pentanol used primarily as a pharmaceutical or pigment solvent.

Using your knowledge of the above reaction, suggest the reagents and conditions you would use in a two-step synthesis of 2-methyl-butan-2-ol from butan-2-ol, identifying the intermediate.

(ii) Suggest the reagents and conditions for a reaction that could be used to distinguish between 2-bromobutane and 2-methyl-butan-2-ol. You should state how each of them react in the test.

[5]

(c) Silicon and beryllium are two elements used as starting reagents in many chemical reactions.

(i) In each of the following reactions, describe the way in which the oxide of the named element is reacting and discuss whether its behaviour is what you would expect from the position of the element in the Periodic Table.

- Silicon: $MgO + \mathbf{Si}O_2 \longrightarrow MgSiO_3$
- Beryllium: $2\ NaOH + \mathbf{Be}O \longrightarrow Na_2BeO_2 + H_2O$

(ii) Explain why BeO shows the property in **(c)(i)**.

[4]

(d) Aluminium sulfate and calcium oxide are sometimes added to water supplies to co-precipitate suspended solids and bacteria. A small amount of aluminium remains in solution and its presence in drinking water may contribute to the mental illness known as Alzheimer's disease.

(i) An insoluble aluminium compound might be formed when water, aluminium sulfate and calcium oxide are mixed. Write a balanced equation, with state symbols, to show its production.

(ii) Suggest why it is important not to add too much calcium oxide.

[2]

[Total: 20]

2 Chemical energy can be directly converted to electrical energy by devices such as batteries and fuel cells.

The nickel-cadmium (Ni-Cd) battery is a type of rechargeable battery used in many portable devices like hand phones and cameras.

The cathode is made of a nickel compound, NiO(OH) and the anode is made of cadmium. The electrolyte is aqueous potassium hydroxide. When the battery discharges, $Ni(OH)_2(s)$ and $Cd(OH)_2(s)$ are formed.

(a) **(i)** Construct the half-equations at each electrode of this Ni-Cd electrochemical cell during the discharging process.

(ii) Given that

	E^o /V
$Cd(OH)_2$ / Cd	–0.89
NiO(OH) / $Ni(OH)_2$	+0.49

Calculate the electromotive force of the cell.

(iii) Ni-Cd batteries can be recharged by applying current across the two electrodes. How long would it take to recharge a Ni-Cd battery at a current of 2.0 A, if 5.6 g of cadmium (A_r = 112) has been converted to $Cd(OH)_2$ during the discharging process?

[5]

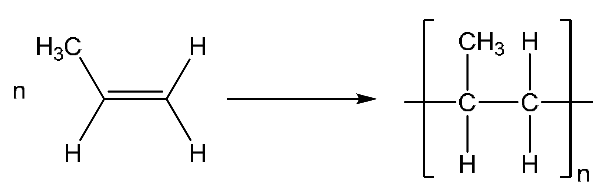
(b) In aqueous solution, transition metal ions such as Ni^{2+} and Fe^{3+} exist as aqua complexes, $M(H_2O)_n^{m+}$.

Using relevant data from the *Data Booklet*, explain the following observations, writing equations where necessary.

When iodide ions were added to aqueous iron(III) chloride, a brown solution was obtained. However, no such observation was seen when the experiment was repeated in the presence of cyanide ions.

[4]

(c) Nickel and its complexes are also used as a catalyst in the polymerisation of alkenes.



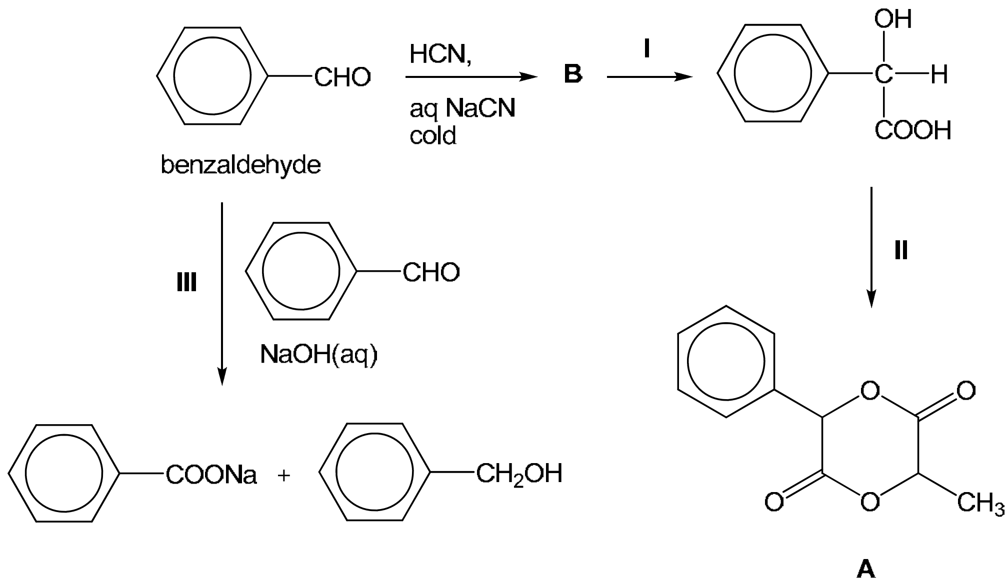
(i) Explain whether the enthalpy change for the polymerisation will be positive or negative.

(ii) Explain the entropy change of this reaction.

(iii) Hence, suggest whether the polymerisation process becomes spontaneous at high or low temperature.

[4]

(d) In the organic reaction scheme below, compound **A** can be formed from benzaldehyde in a series of steps.



(i) Draw the structure of **B**.

(ii) Suggest the type of reaction for steps **II** and **III**.

(iii) Suggest reagents and conditions for steps **I** and **II**.

(iv) Suggest whether propanal or benzaldehyde reacts with HCN at a faster or slower rate.

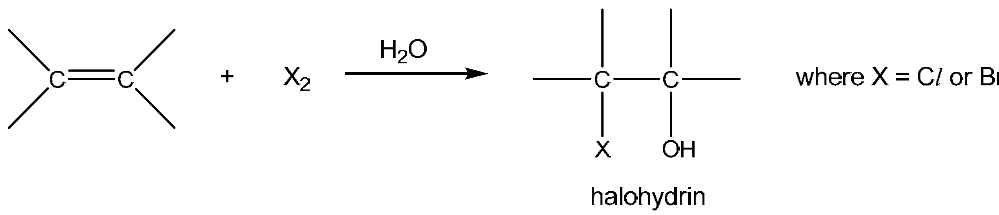
[7]

[Total: 20]

3 **(a)** When 15.0 cm^3 of an alkene **C** underwent complete combustion in 100 cm^3 of excess oxygen, the remaining gases occupied 77.5 cm^3. On shaking these gases with aqueous sodium hydroxide, a reduction in gaseous volume of 45 cm^3 was noted. Determine the molecular formula of compound **C** and hence deduce its identity. All volumes were measured at r.t.p.

[3]

(b) Formation of 1,2-halo alcohols, also known as halohydrins, occurs via the addition reaction between an alkene and a halogen in the presence of water.



In a series of experiments, the reaction between alkene **C** and aqueous bromine was carried out with different concentrations of the two reagents, and the following relative initial rates were obtained.

Experiment	[alkene **C**] / mol dm^{-3}	[Br_2] / mol dm^{-3}	initial rate / mol dm^{-3} s^{-1}
1	0.020	0.020	1.00×10^{-3}
2	0.030	0.020	1.50×10^{-3}
3	0.040	0.030	3.00×10^{-3}

(i) Use these data to deduce the order of reaction with respect to each of the two reagents, showing how you arrive at your answers.

(ii) Hence write a rate equation for the reaction.

(iii) Calculate the rate constant for the reaction, giving its units.

(iv) Why is it not possible to determine the order with respect to water in this experiment?

(v) Comment on how the rate of reaction may change if chlorine is used instead of bromine in the reaction with alkene **C**.

[7]

(c) The mechanism of the addition reaction between but-1-ene and aqueous Br_2 involves three steps.

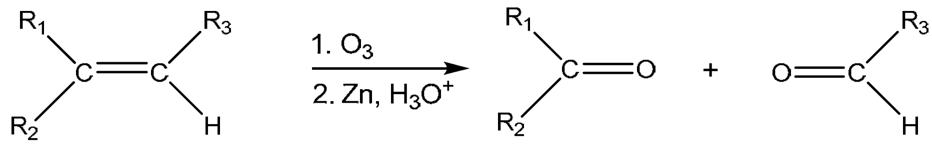
- There is an initial attack by the π electron pair of the alkene on Br_2 to yield a carbocation intermediate.
- This is followed by the nucleophilic attack of the lone pair of electrons on oxygen in water on the carbocation intermediate.
- The third step involves the loss of H^+ ion which then yields the neutral bromohydrin.

Use the information given above to describe a mechanism that is consistent with your rate equation, and indicate which step in the mechanism is the rate determining step.

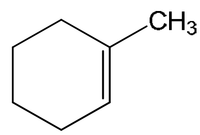
[3]

(d) Apart from hot acidified $KMnO_4$, other powerful oxidising agents such as ozone (O_3) can also cause C=C bond cleavage in alkenes.

In the ozonolysis of alkene, the C=C bond is cleaved and oxygen becomes doubly bonded to each of the original alkene carbons.



Predict the organic products of the following reactions when compound **D** is reacted with



Compound **D**

(i) hot acidified $KMnO_4$

(ii) O_3, followed by Zn and H_3O^+

(iii) IBr in CCl_4

[3]

(e) **(i)** Describe what you see when phosphorus is burned in excess oxygen.

(ii) Describe what you would observe when separate samples of oxides of sodium and phosphorus react with water containing universal indicator. Write equations for the reactions you described.

[4]

[Total: 20]

4 **(a)** In a laboratory setting, amino acid analysis of the peptide angiotensin II shows the presence of eight different amino acids in equimolar amounts: Arg, Asp, His, Ile, Phe, Pro, Tyr, and Val.

Partial hydrolysis of angiotensin II yield the following fragments:

Asp-Arg-Val-Tyr

Ile-His-Pro

Pro-Phe

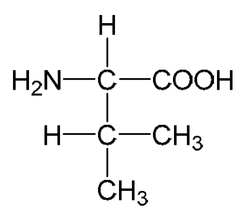
Val-Tyr-Ile-His

Deduce the *primary structure* of angiotensin II.

[2]

(b) **(i)** Explain what is meant by the term *zwitterion.*

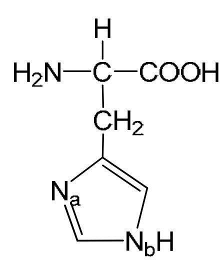
(ii) The amino acid, valine, which is found in angiotensin II, has the following structure.



By using suitable equations, show how the zwitterionic form of valine can act as a buffer.

[3]

(c) The amino acid, histidine has the following structure.

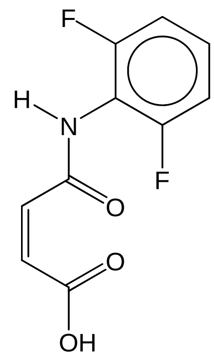


(i) At pH 6, only N_a will be protonated while N_b remains unchanged. Suggest a reason for this.

(ii) Explain, in terms of R-group interaction in the histidine amino acid residue, how a low pH of 2 might affect the tertiary structure of angiotensin II.

[3]

(d) The Ugi reaction is a multi-component reaction in organic chemistry involving an aldehyde and an isocyanide to form a bis-amide. Compound **E** is one such product to be formed from this reaction.



Compound **E**

(i) Describe the conditions needed to hydrolyse amides in the laboratory.

(ii) Predict the products of the hydrolysis of Compound **E**.

(iii) One of the hydrolysed products of Compound **E** in **(d)(ii)** can subsequently react with acidified potassium manganate(VII) to give a gaseous compound **F**.

Identify compound **F.**

[3]

(d) An aqueous solution of diamine, $NH_2CH_2CH_2CH_2NH_2$ has pK_{a1} of 10.6 and pK_{a2} of 8.9.

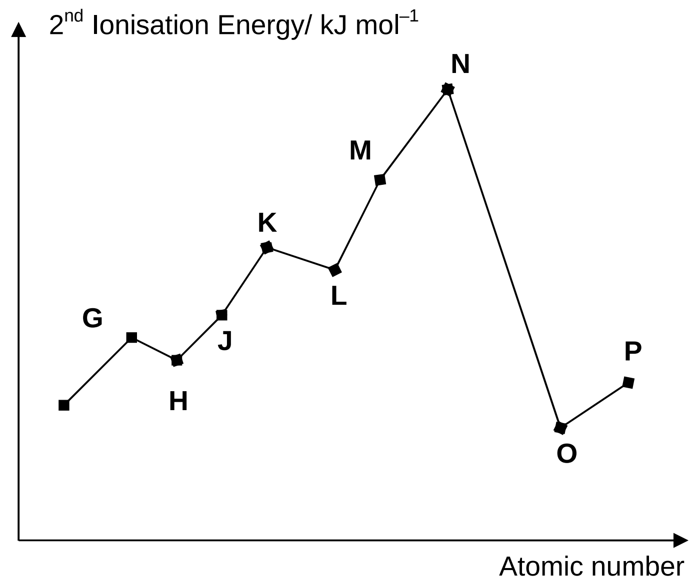
When the aqueous solution is titrated with HC*l*(aq), two successive acid-base reactions occur.

You are given that the initial pH of the aqueous solution of diamine is 11.8.

Sketch the pH changes you would expect to obtain when 30 cm^3 of 0.10 mol dm^{-3} HC*l* (aq) is added to 10 cm^3 of 0.10 mol dm^{-3} solution of the diamine. Include the various key points on the curve. You are not required to calculate the pH at the equivalence points.

[4]

(e) The graph below shows the second ionisation energy of eight elements with consecutive proton number.



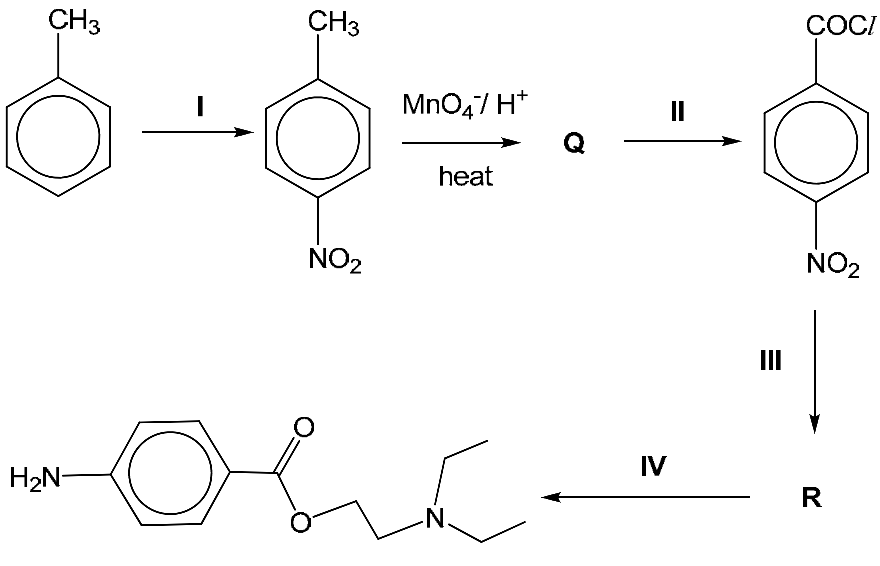
(i) Which of the above element, **G** to **O**, is aluminium? Explain your answer.

(ii) Hence, or otherwise, explain the drop in value from **G** to **H**.

[5]

[Total: 20]

5 Procaine, $C_{13}H_{20}N_2O_2$, a local anaesthetic, can be synthesised through the following reaction pathway.



Procaine

(a) State the reagents and conditions for reactions **I, II, III** and **IV**.

[5]

(b) Draw the structures of compounds **Q** and **R**.

[2]

(c) Suggest and explain how the acidity of compound **Q** might compare with that of benzoic acid.

[2]

(d) In a laboratory preparation of procaine, when 15.4 g of methylbenzene was used, 75 % of the procaine (M_r = 236) was obtained after purification.

Calculate the mass of procaine collected.

[2]

(e) A colourless crystalline solid that was used in textile processing and as an oil and fat preservative, is recently found in starch-based Taiwanese products. This chemical is linked to kidney failure.

This crystalline solid exist as isomers, **S** and **T** with the molecular formula $C_4H_4O_4$. The properties of the two isomers are given below.

- Aqueous solutions of **S** and **T** turn blue litmus red.
- Both **S** and **T** decolourise a solution of bromine in tetrachloromethane.
- When heated, both **S** and **T** undergo dehydration to lose a molecule of water. However, the dehydration of **T** requires a lower temperature than **S**.

(i) Suggest, with reasons, the structural formulae of **S** and **T**.

(ii) State the type of isomerism present in **S** and **T**.

(iii) With reference to your answer in **(ii)**, explain why the dehydration of **T** requires a lower temperature than **S**.

(iv) Write an equation to show the dehydration of **T**.

[6]

(f) The monocarboxylate ion of a dicarboxylic acid, $HO_2C–(CH_2)_n–CO_2^-$, is an amphiprotic species.

(i) Suggest what is meant by an *amphiprotic* species.

(ii) Using malonic acid, $HO_2CCH_2CO_2H$ as an example, write two equations to show how its **monocarboxylate ion** can act as an amphiprotic species.

[3]

[Total: 20]